

Recursion

1) reverse an array using recursion

keep pointer i at start =0 and n-i-1 points to ends

swap these start and end

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

```
void revarray(int i, int arr[], int n)
```

```
{
```

```
    if (i >= n / 2)
```

```
        return;
```

```
    swap(arr[i], arr[n - i - 1]);
```

```
    revarray(i + 1, arr, n);
```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    cout << "Enter value of n: ";
```

```
    cin >> n;
```

```
    int a[n];
```

```
    cout << "Enter value of arr: ";
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        cin >> a[i];
```

```
    }
```

```
    cout << endl
```

```
        << "Reversed array is: ";
```

```
    revarray(0, a, n);
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        cout << a[i] << " ";
```

```
    }
```

```
    return 0;
```

```
}
```

2) check string is pallindrome or not

```
#include <bits/stdc++.h>
using namespace std;

bool pallindrome(int i, string &s)
{
    if (i >= s.size() / 2)
        return true;
    if (s[i] != s[s.size() - i - 1])
        return false;
    return pallindrome(i + 1, s);
}

int main()
{
    string str = "madam";
    if (pallindrome(0, str))
        cout << "Yes";
    else
        cout << "No";
}
```

3) multiple recursion problem (fibonachi numbers)

```
int f(int n)
{
    if (n <= 1)
        return n;
    int last = f(n - 1);
    int slast = f(n - 2);
    return last + slast;
}
```